

Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
CPT304		CPT	

2nd SEMESTER 2022/23 Final EXAMINATIONS

BACHELOR DEGREE – Year 4

Software Engineering II

TIME ALLOWED: 2 Hours

INSTRUCTIONS TO CANDIDATES

- 1、 This is a closed-book examination, which is to be written without books or notes.**
- 2、 Total marks available are 100. This will count for 80% in the final marks.**
- 3、 This exam consists of nine (9) questions printed on 4 pages of papers including the cover page.**
- 4、 Answer all questions. There is NO penalty for providing a wrong answer.**
- 5、 Answers should be written in the answer booklet(s) provided.**
- 6、 Only English solutions are accepted.**
- 7、 All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.**

Answer all questions. Write your answers in the provided answer booklet.

Question 1

In the article, No Silver Bullet: Essence and Accidents of Software Engineering written by Frederick P. Brooks, the author claimed that one of the very promising attacks on the essence difficulties in the software process is the requirements refinement and rapid prototyping. Explain why requirements refinement and rapid prototyping are claimed to be promising attacks. (18 marks)

Question 2

Cohesion and coupling are common concepts in designing modular software systems.

1. Explain the concepts of cohesion and coupling. (6 marks)
2. The User class shown in Figure 1 below is considered a class with low cohesion, explain. (4 marks)

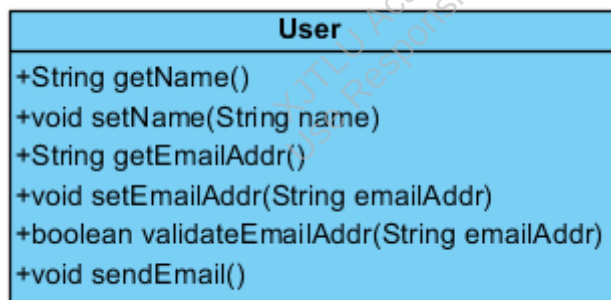


Figure 1

Question 3

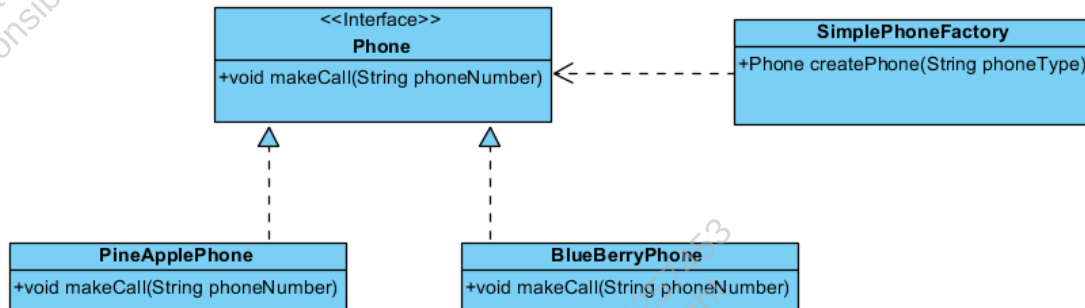
1. Explain the concept of the Open/Closed principle in Object-Oriented design. (4 marks)
2. What is the main objective of the Open/Closed principle? (4 marks)
3. What could be the problem if we achieve the Open/Closed principle using inheritance? (4 marks)

Question 4

In Object Oriented programming, we could use composition instead of inheritance in some situations. What are the THREE advantages of using composition over inheritance? (6 marks)

Question 5

Figure 2 illustrates the class diagram for a simple factory pattern of the Make Phone Call program. The corresponding code fragments are shown in Figure 3 below.

**Figure 2**

```

1 public interface Phone {
2     public void makeCall(String phoneNumber);
3 }

1 public class BlueBerryPhone implements Phone{
2     @Override
3     public void makeCall(String phoneNumber) {
4         // Add BlueBerry specific operation to make call
5     }

1 public class PineApplePhone implements Phone{
2     @Override
3     public void makeCall(String phoneNumber) {
4         // Add PineApple specific operation to make call
5     }

1 public class SimplePhoneFactory {
2     public Phone createPhone(String phoneType){
3         if (phoneType.equals(anObject:"BlueBerry")){
4             return new BlueBerryPhone();
5         } else {
6             return new PineApplePhone();
7         }
8     }
9 }
  
```

Figure 3

1. What is the main problem of the simple factory pattern? (6 marks)
2. Modify the class diagram shown in Figure 2 to provide a solution for the main problem of the simple factory pattern. (8 marks)

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Question 6

List and explain THREE benefits of software reuse.

(12 marks)

Question 7

Compared to the Object-Oriented approach, how does the software component approach improve software reusability?

(12 marks)

Question 8

Explain the term 'scalability' in the context of system architecture. Clarify the differences between 'scaling up' and 'scaling out'.

(10 marks)

Question 9

One of the reasons an organization adopts Software as a Service (SaaS) is the potential for a reduction in IT support costs. Explain THREE ways the Software as a Service strategy could reduce the IT support cost.

(6 marks)

The end of the paper